

analysed. Why? Because Big Data is really hard to manage. Before we move on to discussing the open source tools that can help us manage Big Data, it is necessary to understand why we need these tools in the first place.

Why is Big Data so hard to handle?

1. Almost 90 per cent of the Big Data we produce today is unstructured, and there are no predefined models that it adheres to. And there is no quick and easy way to access and analyse unstructured data.
2. There is an exponential shortage of data analysts in the market. We have the data, but we do not have enough analysts that can come up with clever ways of handling it.
3. Not only is Big Data unstructured, it is also extremely complex. This means traversing through it requires advanced algorithms that are being run on top of grade-A machinery, and not everyone can afford that.
4. Scalability is a major ongoing challenge with Big Data. What is the best, most optimal method of storing data of such magnitude?
5. We have multiple sources from which we accrue data, and there are no clean ways to integrate disparate data from so many sources. This poses a huge challenge for companies.
6. With Big Data come big mistakes. Storing massive and complex data in a secure manner is a really daunting challenge. It is also the need of the hour. Since the beginning of the pandemic, more people are stuck at home with their computers than ever before, and cybercrime and data theft have gone up by 400 per cent.

I realise that Big Data poses social, economic, environmental and privacy related challenges as well. But since our focus on this article is to stick to the technical side of things, those are the issues I shall focus on.

The role of open source software

There is a reason why the Mars Rover runs on Linux. Open source has been the single biggest reason for the rapid advancement of computer technology. The beauty of open source is that it is voluntary. In the open source community, you get to select and work on projects that matter to you. You try to solve problems that you face. The challenges that we face with Big Data are far greater and way more complex for some random company to solve with a proprietary piece of software. It requires the collective genius of people belonging to different aspects of the industry.

Just as open source has been the primary reason for the Big Data revolution, so has it been at the forefront of developing tools to counter the challenges that we face with this data. And while there are hundreds of popular open source projects that are focused on Big Data management, in this article, we shall focus on the top three that caught my attention. We shall discuss the Hadoop framework, the Spark analytics engine and the Kafka framework.

Hadoop

Developed by the acclaimed Apache Foundation, Hadoop is a framework for Big Data management. It uses a

distributed systems architecture, which has the ability to store and easily access vast amounts of data. Major companies such as British Airways, Marks and Spencer, Royal Mail and Expedia use Hadoop.

While there are many different types of distributed systems solutions for Big Data, what makes Hadoop stand out are three unique components, namely, HDFS, MapReduce and YARN.

HDFS

Every distributed storage system requires a mechanism for storing and accessing data. The Hadoop Distributed File System (HDFS) is a powerful and dynamic tool that is efficient at handling Big Data. HDFS breaks down data into blocks, as shown in Figure 2. The default size of each block is 128MB. Each block of data is stored in a different computer. This also makes accessing the data simple, as the HDFS only has to access the specific block of data that the user is searching for. HDFS is a resilient framework, thanks to its replication methodology. Not only is data divided into blocks, but every single block is replicated by a factor of three into multiple other computers. This ensures that if a computer goes offline, the data stored within it can still be accessed from other computers.

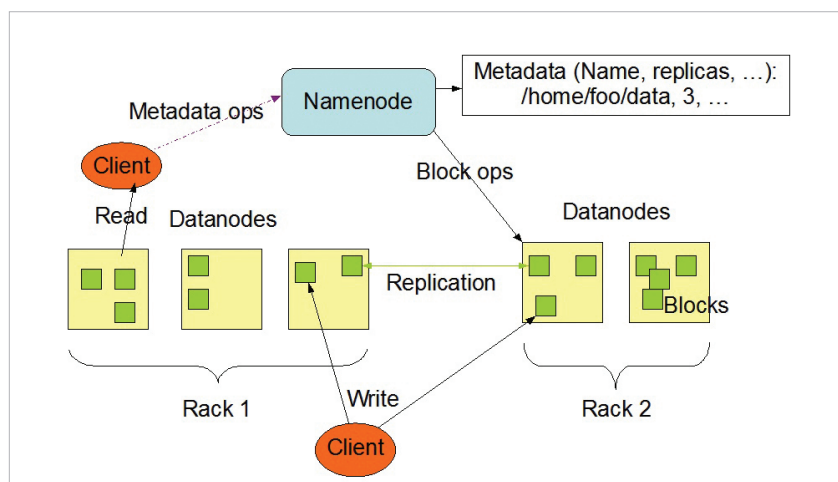


Figure 2: HDFS architecture (Courtesy: Apache Hadoop)